



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2009

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

Examiner: B. Weiss

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☒ Mr Barrett
- ☐ Mr Fletcher
- ☐ Mr Weiss

Ben L

Student Number 194 88152

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	Total	Total
Mark	$\frac{11}{15}$	$\frac{14}{15}$	$\frac{13}{15}$	$\frac{14}{15}$	$\frac{12}{15}$	$\frac{15}{15}$	$\frac{8}{15}$	$\frac{15}{15}$	$\frac{102}{120}$	$\frac{100}{100}$

Question 1

(a) Find the following integrals:

(i) $\int \tan^3 x \, dx$ 3

(ii) $\int \frac{dx}{x^2 - 6x + 13}$ 2

(b) Evaluate

(i) $\int_0^1 \frac{x}{\sqrt{4-x^2}} \, dx$ 3

(ii) $\int_0^{\frac{\pi}{2}} x \cos^2 x \, dx$ 3

(c) (i) Show that if $I_n = \int_0^1 x^n e^{-x} \, dx$, then $I_n = n \cdot I_{n-1} - \frac{1}{e}$ 2

(ii) Hence find $\int_0^1 x^3 e^{-x} \, dx$. 2

Question 2 (Start a new page)

(a) Find $\sqrt{6i-8}$, and hence solve the equation $2z^2 - (3+i)z + 2 = 0$. 4

(b) Solve $3x^3 - 10x^2 + 7x + 10 = 0$ given that $x = 2 - i$ is a root of the equation. 3

(c) The polynomial equation $P(x) = x^3 + px^2 + q = 0$ has roots α , β and γ .
Form the polynomial equation with roots given by

(i) $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$ 2

(ii) α^2 , β^2 and γ^2 2

(d) Use the method of cylindrical shells to find the volume of the solid generated by rotating the region bounded by $y = \ln x$, the x -axis and the lines $x = 1$ and $x = e$, about the y -axis. 4

Question 3 (Start a new page)

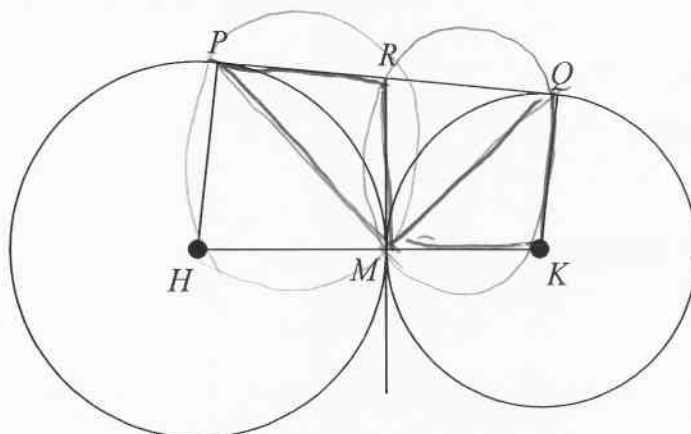
- (a) (i) Prove that the equation of the tangent at the point $\left(t, \frac{1}{t}\right)$ to the hyperbola $xy = 1$ is $x + t^2y = 2t$. 2

- (ii) The tangent at a point P on the hyperbola $xy = 1$ meets the y -axis at A , and the normal at P meets the x -axis at B . Find the equation of the locus of the midpoint of AB as P moves on the hyperbola. (Draw a diagram) 3

- (b) $P(a \cos \alpha, b \sin \alpha)$ and $Q(a \cos \beta, b \sin \beta)$ are the endpoints of a focal chord of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. 3

Show that $e = \frac{\sin(\alpha - \beta)}{\sin \alpha - \sin \beta}$.

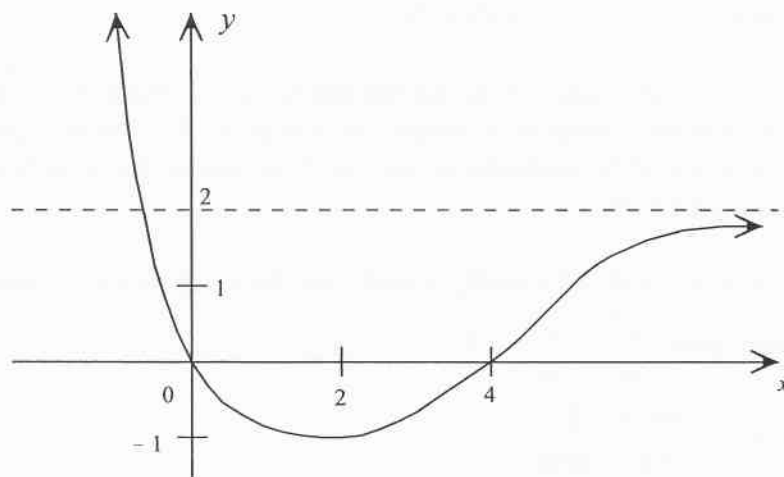
- (c) Shown below are two circles with centres H and K which touch at M . PQ and RM are common tangents.



- (i) Show that quadrilaterals $HPRM$ and $MRQK$ are cyclic. 2
- (ii) Prove that triangles PRM and MKQ are similar. 2
- (d) Show that the polynomial equation $4x^3 + 20x^2 - 23x + 6 = 0$ has a double root, and find the value of each of its roots. 3

Question 4

- (a) The diagram shows the graph of $y = f(x)$.



Sketch on separate diagrams, the following curves, indicating clearly any turning points and asymptotes.

(i) $y = \frac{1}{f(x)}$ 1

(ii) $y = [f(x)]^2$ 2

Draw neat sketches of the following:

(b) $y = x \sin x$ 2

(c) $y = \sin^{-1}(\sin x)$ 2

(d) $y = x^2 - \frac{1}{x}$ 3

(e) (i) $f(x) = \frac{x^2 - 4}{x - 3}$ 3

(ii) $[f(x)]^2 = \frac{x^2 - 4}{x - 3}$ 2

Question 5 (Start a new page)

- (a) (i) Express the complex number $z = -\sqrt{3} + i$ in mod-arg form. 1
- (ii) Hence, or otherwise, show that $z^7 + 64z = 0$. 2
- (b) Find the equation, in Cartesian form, of the locus of the point z if
$$\operatorname{Re} \left[\frac{z-4}{z} \right] = 0$$
 3
- (c) Sketch the region S in the complex plane, where
$$S = \left\{ |z| \leq 1 \quad \text{and} \quad 0 \leq \arg z < \frac{\pi}{3} \right\}$$
 2
- (d) (i) Use de Moivre's theorem to express $\cos 5\theta$ and $\sin 5\theta$ in terms of $\sin \theta$ and $\cos \theta$. 3
- (ii) Hence express $\tan 5\theta$ as a rational function of t , where $t = \tan \theta$. 2
- (iii) Find $\tan \frac{\pi}{5} \cdot \tan \frac{2\pi}{5} \cdot \tan \frac{3\pi}{5} \cdot \tan \frac{4\pi}{5}$ 2

Question 6 (Start a new page)

- (a) A particle of mass 1 kg is projected upwards with initial speed 10 ms^{-1} . 5
The air resistance is given by $R = \frac{1}{10} v^2$.
Take the acceleration due to gravity to be 10 ms^{-2} .
Find the maximum height reached, and the time taken to reach this height.
- (b) Find the largest coefficient in the expansion of $(2x + 3)^{21}$. 3
- (c) If $x^m y^n = k$, where k is a constant, show that $\frac{dy}{dx} = -\frac{my}{nx}$. 3
- (d) Use the expansion of $(1+x)^{2n}$ to show that
- (i)
$$\binom{2n}{1} + \binom{2n}{2} + \binom{2n}{3} + \dots + \binom{2n}{2n} = 4^n - 1$$
 2
- (ii) Use the identity $(1+x)^{2n} \equiv (1+x)^n (1+x)^n$ to show that
$$\binom{2n}{2} = 2 \cdot \binom{n}{2} + \binom{n}{1}^2$$
 2

Question 7 (Start a new page)

- (a) Use the process of mathematical induction to show that

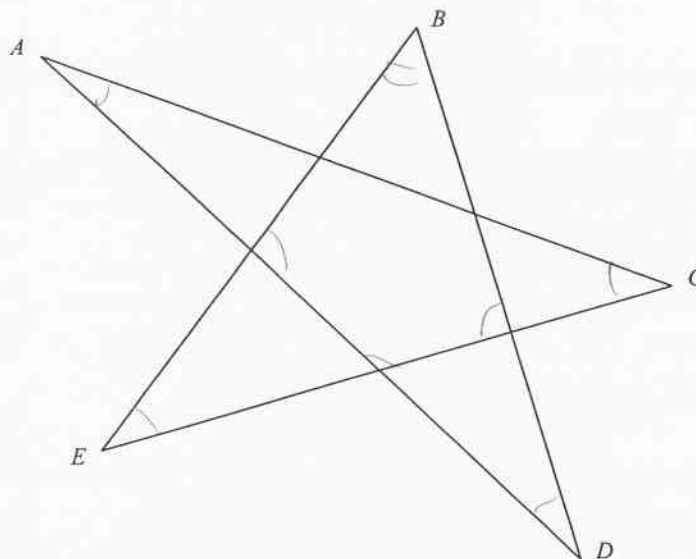
$$\sum_{k=1}^n \frac{k}{(k+1)!} = 1 - \frac{1}{(n+1)!} \quad 5$$

- (b) With the aid of a diagram, show that the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is given by } A = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx.$$

Hence show that the area of the ellipse is πab .

- (c)



Prove that $\angle A + \angle B + \angle C + \angle D + \angle E = 180^\circ$

- (d) The base of a solid is the region bounded by the parabolas $x = y^2$ and $x = 4 - 3y^2$, and the cross-sections perpendicular to the x -axis are squares.

- (i) Draw a neat sketch of this solid.

- (ii) Find the volume of the solid.

$$\begin{aligned} 3y^2 &= 4 - y^2 & 1 \\ y^2 &= \frac{4-y^2}{3} & 3 \\ y &= \sqrt{\frac{4-y^2}{3}} \end{aligned}$$

Question 8 (Start a new page)

(a) If a and b are positive numbers such that $a + b = 1$, prove that

(i) $a + b \geq 2\sqrt{ab}$

1

(ii) $\frac{1}{a} + \frac{1}{b} \geq 4$

2

(iii) $a^2 + b^2 \geq \frac{1}{2}$

2

(iv) $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right) \geq 9$

2

(b) A particle is projected from ground level so that it just clears two poles of height h at distances of b and c metres from the point of projection. If v m/s is the velocity of projection, and θ is the angle of projection to the horizontal:

(i) Show that $y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$

2

(ii) Show that $v^2 = \frac{(b+c)g \sec^2 \theta}{2 \tan \theta}$

3

(iii) Hence or otherwise show that $\tan \theta = \frac{h(b+c)}{bc}$

3

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Q1) a) i/ $\int \tan^3 x \, dx$

$$= \int \tan x (\sec^2 x - 1) \, dx$$

$$= \frac{\tan^2 x}{2} - \log |\sec x| + C$$

ii/ $\int \frac{dx}{x^2 - 6x + 13}$

$$= \int \frac{dx}{(x-3)^2 + 4}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C$$

b) i/ $\int_0^1 \frac{x}{\sqrt{4-x^2}} \, dx$

Let $u = 4-x^2$
 $du = -2x \, dx$

$$= -\frac{1}{2} \int_3^4 u^{-1/2} \, du$$

$$= -\frac{1}{2} \int_3^4 u^{-1/2} \, du$$

$$= \frac{1}{2} \left[2u^{1/2} \right]_3^4$$

$$= 2 - \sqrt{3}$$

ii/ $\int_0^{\pi/4} \frac{1}{x} \cos^2 x \, dx$

$$= \int_0^{\pi/4} x \left(\frac{1}{2} + \frac{\cos 2x}{2} \right) dx$$

$$= \frac{x^2}{4} \Big|_0^{\pi/4} + \int_0^{\pi/4} x \frac{\cos 2x}{2} \, dx$$

$$= \frac{\pi^2}{16} + x \frac{\sin 2x}{4} \Big|_0^{\pi/4} - \int_0^{\pi/4} \frac{\sin 2x}{4} \, dx$$

$$= \frac{\pi^2}{16} + \frac{\cos 2x}{8} \Big|_0^{\pi/4}$$

$$= \frac{\pi^2}{16} + \frac{-1}{8} - \frac{1}{8}$$

c) i/ $I_n = \int_0^1 x^n e^{-x} \, dx$

$$= -x^n e^{-x} \Big|_0^1 + \int_0^1 n x^{n-1} e^{-x} \, dx$$

$$= -e^{-1} + n \int_0^1 x^{n-1} e^{-x} \, dx$$

$$\therefore I_n = n I_{n-1} - \frac{1}{e}$$

ii/ $I_0 = \int_0^1 e^{-x} \, dx$

$$= -\frac{1}{e^x} \Big|_0^1$$

$$= 1 - \frac{1}{e}$$

$$I_1 = I_0 - \frac{1}{e}$$

$$= 1 - \frac{1}{e} - \frac{1}{e}$$

$$= 1 - \frac{2}{e}$$

$$I_2 = 2 I_1 - \frac{1}{e}$$

$$= 2 \left(1 - \frac{2}{e} \right) - \frac{1}{e}$$

$$= 2 - \frac{5}{e}$$

$$I_3 = 3 I_2 - \frac{1}{e}$$

$$= 3 \left(2 - \frac{5}{e} \right) - \frac{1}{e}$$

$$= 6 - \frac{16}{e}$$

$$\therefore \int_0^1 x^3 e^{-x} \, dx = 6 - \frac{16}{e}$$

Q2) a) $\sqrt{ci-8} = a+ib$

$$a^2 - b^2 = -8$$

$$2ab = 6$$

$$a = \frac{b}{3}$$

$$\frac{b^2}{9} - b^2 = -8$$

$$-8b^2 = 8$$

$$\frac{b^2}{9} = 8$$

$$b = \pm 3$$

$$\therefore a = \pm 1$$

$$2z^2 - (3+i)z + 2 = 0$$

$$z = \frac{(3+i) \pm \sqrt{9+6i-1-4.2.2}}{4}$$

$$= \frac{3+i \pm \sqrt{6i-8}}{4}$$

$$= \frac{3+i \pm (1+3i)}{4}$$

$$= 1+i \text{ or } \frac{1-i}{2}$$

b) If $2-i$ is a root so is $2+i$

$$(x-(2+i))(x-(2-i))$$

$$= x^2 - (2-i)x - (2+i)x + 5$$

$$= x^2 - 4x + 5$$

$$x^2 - 4x + 5 \quad \begin{array}{l} 3x + 2 \\ 3x^2 - 10x^2 + 7x + 10 \\ 3x^3 - 12x^2 + 15x \\ 2x^2 - 8x + 10 \end{array}$$

$$\therefore \text{roots are } x = (2-i), (2+i), -\frac{2}{3}$$

c) $P(x) = x^3 + px^2 + q = 0$

$$P\left(\frac{1}{x}\right) = \frac{1}{x^3} + \frac{p}{x} + q = 0$$

$$= 1 + px^2 + qx^3 = 0$$

$$\therefore qx^3 + px^2 + 1 = 0$$

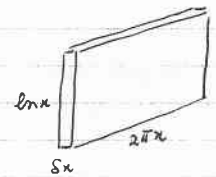
ii) $P(x) = x^3 + px^2 + q = 0$

$$P\left(\frac{1}{x}\right) = x^3 + px^2 + q = 0$$

$$x^3 = -(px^2 + q)$$

$$x^3 = p^2 x^2 + 2pqx + q^2$$

$$\therefore x^3 - p^2 x^2 - 2pqx - q^2 = 0$$



Volume of slice $2\pi x \cdot 8x \cdot \ln x$

$$\text{Volume} = 2\pi \int_1^e x \ln x \, dx$$

$$= 2\pi \left[\frac{x^2}{2} \ln x \right]_1^e - \frac{1.2\pi}{2} \int_1^e x \, dx$$

$$= 2\pi \left[\frac{e^2}{2} \right] - \pi \left[\frac{x^2}{2} \right]_1^e$$

$$= \pi e^2 - \frac{\pi e^2}{2} + \frac{\pi}{2}$$

$$= \frac{\pi}{2} (e^2 + 1) \omega^3$$

Q3) a) i/ $y = \frac{1}{x}$

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

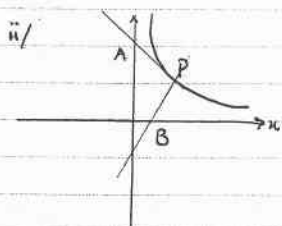
at $x = t$ $\frac{dy}{dx} = -\frac{1}{t^2}$

$\therefore$ eqn of tangent

$$y - \frac{1}{t} = -\frac{1}{t^2}(x - t)$$

$$t^2 y - t = -x + t$$

$$x + t^2 y = 2t$$



tangent meets y axis at $(0, \frac{2}{t})$

normal meets x axis at

$$t^2 x = y + t^3 - \frac{1}{t}$$

$$(t - \frac{1}{t^3}, 0)$$

Mid point $(\frac{t^4 - 1}{2t^3}, \frac{1}{t})$

$$y = \frac{1}{t}$$

$$x = \frac{t^4 - 1}{2t^3}$$

$$= \frac{\frac{1}{t^4} - 1}{\frac{2}{t^3}}$$

$$x = \frac{1 - y^4}{2y}$$

d) $P(x) = 4x^3 + 20x^2 - 23x + 6$

$$P'(x) = 12x^2 + 40x - 23$$

$$= (6x + 23)(2x - 1)$$

$$\therefore (6x + 23)(2x - 1) = 0$$

$$\text{when } x = \frac{1}{2} \text{ or } -\frac{23}{6}$$

$$P(\frac{1}{2}) = 0 = P'(\frac{1}{2})$$

$\therefore x = \frac{1}{2}$ is the double root

Using product of roots

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \alpha = -\frac{6}{4}$$

$$\alpha = -6$$

$\therefore$ roots are $\frac{1}{2}, \frac{1}{2}, -6$

Q3)

b)

$$\text{grad of } m = \frac{b(\sin \alpha - \sin \beta)}{a(\cos \alpha - \cos \beta)}$$

eqn of PQ

$$y - b \sin \alpha = \frac{b}{a} \left(\frac{\sin \alpha - \sin \beta}{\cos \alpha - \cos \beta} \right) (x - a \cos \alpha)$$

Now focal chord $\Rightarrow (ae, 0)$

satisfies eqn

$$-b \sin \alpha = \frac{b(\sin \alpha - \sin \beta)}{a(\cos \alpha - \cos \beta)} (ae - a \cos \alpha)$$

$$\therefore -\sin \alpha (\cos \alpha - \cos \beta) = e - \cos \alpha$$

$$-(\sin \alpha \cos \alpha - \sin \alpha \cos \beta) + \cos \alpha = e$$

$$e = \frac{-\sin \alpha \cos \alpha + \sin \alpha \cos \beta + \cos \alpha}{\sin \alpha - \sin \beta}$$

$$= \frac{\sin(\alpha - \beta)}{\sin \alpha - \sin \beta}$$

c) i/ In HPRM

$\angle HPR = \angle RPH$ (angle between radius & tangent)

$\therefore$ HPRM is a cyclic quad (opp angles add to 180°)

Similarly MRQK

ii/ Join PM and QM

In ΔPRM and ΔMQK

$\angle PRM = \angle MKQ$ (ext angle of a cyclic quad)

$PR = RM$ (tangents from an ext pt)

$KM = KQ$ (radii)

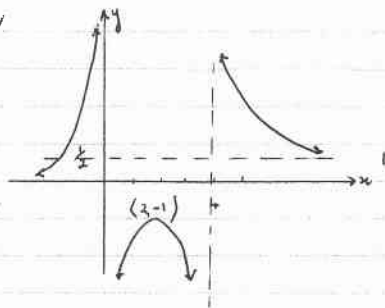
Δ isos

$\angle RPM = \angle RMP = \angle KMQ$

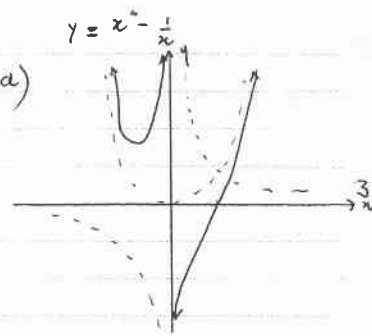
$\therefore \Delta PRM \cong \Delta MQK$ (equilateral)

Q4)

a) i/

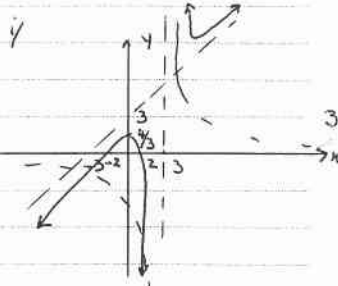


d)

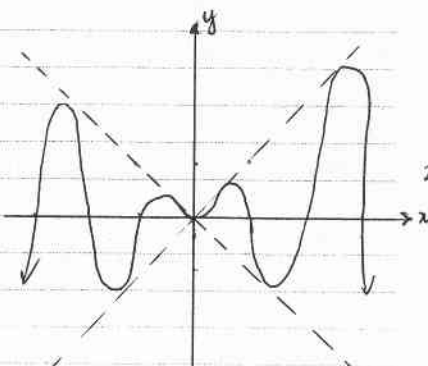


c) $f(x) = \frac{x^2 - 4}{x - 3}$

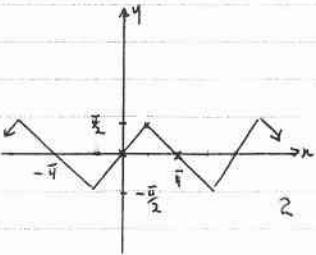
$= x + 3 + \frac{5}{x - 3}$



b)

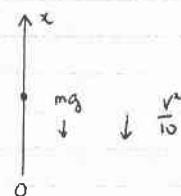


c)



Q6)

a)



$m\ddot{x} = -mg - \frac{v^2}{10}$

$\ddot{x} = -g - \frac{v^2}{10} \quad (\because m=1)$

$\frac{dv}{dt} = -g - \frac{v^2}{10}$

$\therefore t = \int \frac{dv}{-g - \frac{v^2}{10}} = -10 \int \frac{dv}{100 + v^2} = -\tan^{-1}\left(\frac{v}{10}\right) + C_1$

when $v=10, t=0 \Rightarrow C_1 = \frac{\pi}{4}$

$\therefore \tan^{-1}\left(\frac{v}{10}\right) = \frac{\pi}{4} - t$

$\frac{v}{10} = \tan\left(\frac{\pi}{4} - t\right)$

when $v=0, t = \frac{\pi}{4}$

$\therefore x = \int 10 \tan\left(\frac{\pi}{4} - t\right) dt = 10 \ln \cos\left(\frac{\pi}{4} - t\right) + C_2$

when $t=0, x=0$

$\therefore C_2 = -10 \ln \frac{1}{\sqrt{2}} = 5 \ln 2$

$\therefore$ when $x = \frac{\pi}{4}, x = 5 \ln 2$ metres.

b)

$(2x+3)^{21}$

$\frac{T_{r+1}}{T_r} = \frac{{}^{21}C_r (2x)^{21-r} 3^r}{{}^{21}C_{r-1} (2x)^{21-r-1} 3^{r-1}} = \frac{{}^{21}C_r}{{}^{21}C_{r-1}} \times 3 = \frac{21!}{r!(21-r)!} \times 3 = \frac{21!}{r!(22-r)!} \times 3$

Now $\frac{66-3r}{2r} \geq 1 \Rightarrow 66-3r \geq 2r \Rightarrow r \leq 13 \frac{1}{2} \Rightarrow r = 13$
 $\therefore T_{14} = \frac{21!}{C_{13}} 2^8 3^{13}$

c)

$x^m y^n = k$

$x^m n y^{n-1} \frac{dy}{dx} + m x^{m-1} y^n = 0$

$x^m n y^{n-1} \frac{dy}{dx} = -m x^{m-1} y^n$

$\therefore \frac{dy}{dx} = \frac{-m x^{m-1} y^n}{n x^m y^{n-1}} = \frac{-m y}{n x}$

i) $(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + \dots + {}^{2n}C_{2n} x^{2n}$

when $x=1$

$2^{2n} = 1 + {}^{2n}C_1 + \dots + {}^{2n}C_{2n}$

$\therefore 4^n - 1 = {}^{2n}C_1 + \dots + {}^{2n}C_{2n}$

ii) $(1+x)^{2n} = (1+x)^n (1+x)^n$

Term in x^2 on RHS

$= {}^nC_2 x^2 + {}^nC_1 x \cdot {}^nC_1 x + {}^nC_0$

equating coeff of x^2

$2n \dots$

Q5) a) i) $z = -\sqrt{3} + i$
 $= 2 \cos\left(\frac{5\pi}{6}\right)$

ii) $z^7 = 2^7 \cos\left(\frac{35\pi}{6}\right)$
 $= 128 \left(\cos\frac{-\pi}{6} + i \sin\frac{-\pi}{6} \right)$
 $= 128 \left(\cos\frac{\pi}{6} - i \sin\frac{\pi}{6} \right)$

$64z = 2.64 \cos\frac{5\pi}{6}$
 $= 128 \left(\cos\frac{5\pi}{6} + i \sin\frac{5\pi}{6} \right)$
 $= 128 \left(-\cos\frac{\pi}{6} + i \sin\frac{\pi}{6} \right)$
 $= -128 \cos\frac{\pi}{6} + 128 i \sin\frac{\pi}{6}$

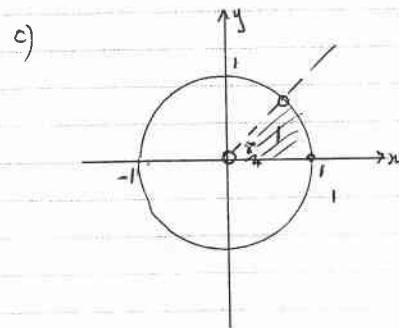
$\therefore z^2 + 64z = 0$

b) $\operatorname{Re}\left(\frac{z-4}{z}\right) = 0$

$\frac{z-4}{z} = \frac{x-4+iy}{x+iy} \times \frac{x-iy}{x-iy}$
 $= \frac{x^2-4x+iy^2+i(xy-xy+4)}{x^2+y^2}$

$\therefore x^2-4x+y^2=0$
 $(x-2)^2+y^2=4$

circle centre (2,0) $r=2$



Q5)

d) i) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$
 $= \cos^5 \theta + 5i \cos^4 \theta \sin \theta + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3$
 $+ 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5$
 $= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta$
 $+ 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$

$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$
 $\sin 5\theta = 5 \sin \theta \cos^4 \theta - 10 \sin^3 \theta \cos^2 \theta + \sin^5 \theta$

ii) Now $\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} = \frac{5C^4S - 10C^2S^3 + S^5}{C^5 - 10C^3S^2 + 5S^5C} \div C^5$
 $= \frac{5t^4 - 10t^2 + t^5}{1 - 10t^2 + 5t^4}$ Let $\tan \theta = t$

iii)

Let $\tan 5\theta = 0$

Then $5\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi, \dots$

$\therefore \theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi, \dots$

$\tan 5\theta = 0$ has an infinite no. of solutions

but $\frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4} = 0$

has only 5 roots

$t(5 - 10t^2 + t^4) = 0$

$\tan \theta = 0$

$\tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}$

product of roots is 5

$\tan \frac{\pi}{5} \cdot \tan \frac{2\pi}{5} \cdot \tan \frac{3\pi}{5} \cdot \tan \frac{4\pi}{5} = 5$

Q7) a) $\sum_{k=1}^n \frac{k}{(k+1)!} = 1 - \frac{1}{(n+1)!}$

Step 1 Let $n=1$

L.H.S. = $\frac{1}{2!} = \frac{1}{2}$ R.H.S. = $1 - \frac{1}{2!} = \frac{1}{2}$

$\therefore$ true for $n=1$

Step 2. Assume true for $n=k$

ie $9 -$

Prove true for $n=k+1$

$\sum_{k=1}^{n+1} \frac{k}{(k+1)!} = 1 - \frac{1}{(n+1)!} + \frac{n+1}{(n+2)!}$

$= 1 - \frac{n+2}{(n+2)!} + \frac{n+1}{(n+2)!}$

$= 1 - \frac{n+2-n-1}{(n+2)!}$

$= 1 - \frac{1}{(n+2)!}$

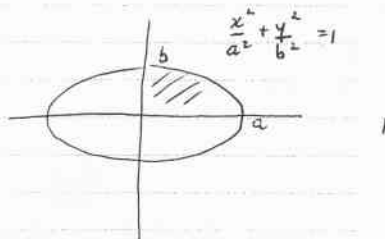
$= \text{R.H.S.}$

So if true for n it is true for $n+1$

Step 3 It is true for $n=1$ and as it

works for n and $n+1$ then it is true by math induction.

b)



Area of ellipse = $4 \times$ shaded area

$A = 4 \int_0^a y \, dx$

$= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$

as $y^2 = \frac{b^2}{a^2} (a^2 - x^2)$

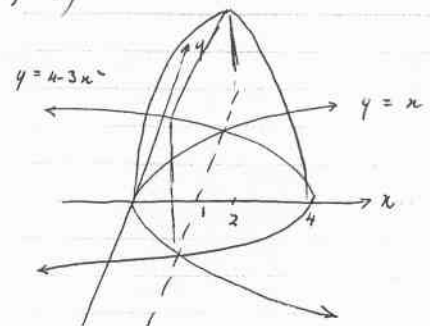
$y = \frac{b}{a} \sqrt{a^2 - x^2}$

$A = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx$

$= \frac{4b}{a} \times \frac{\pi a^2}{4} \left(\frac{1}{4} \text{ of circle} \right)$

$= \pi ab$

Q7) d)



Curves meet at $x=1$

From $x=0$ to 1 the cross-section with co-ord x

has $A = (2y)^2 = 4y^2 = 4x$

$\therefore V = \int_0^1 4x \, dx = 2$

from $x=1$ to 4 the cross section has co-ord x

has $A = (2y)^2 = 4y^2 = \frac{4}{3} (4-x)$

$\therefore V = \frac{4}{3} \int_1^4 (4-x) \, dx = \frac{4}{3} \left[4x - \frac{x^2}{2} \right]_1^4 = \frac{4}{3} \left(16 - 8 - 4 + \frac{1}{2} \right) = 6$

$\therefore$ total volume is $8 \, u^3$

Q8) a) ii/ $\frac{1}{a} + \frac{1}{b} = \frac{b+a}{ab}$

as $a+b=1$

$1 = a+b \geq 2\sqrt{ab}$

$\frac{1}{2} \geq \sqrt{ab}$

$ab \leq \frac{1}{4}$

$\therefore \frac{1}{ab} \geq 4$

$a+b \geq 4$

iii/ $a^2 + b^2 = a^2 + (1-a)^2$
 $= 2a^2 - 2a + 1$
 $= 2(a - \frac{1}{2})^2 + \frac{1}{2} \geq \frac{1}{2}$

equality holds when $a = \frac{1}{2}$

iv/ $(1 + \frac{1}{a})(1 + \frac{1}{b}) = (1 + \frac{1}{a})(1 + \frac{1}{1-a})$
 $= 1 + \frac{1}{a} + \frac{1}{1-a} + \frac{1}{a(1-a)}$
 $= 1 + \frac{2}{a(1-a)}$

Now $a(1-a) = -a^2 + a = -(a - \frac{1}{2})^2 + \frac{1}{4} \leq \frac{1}{4}$

and $1 + \frac{2}{a(1-a)} \geq 1 + \frac{2}{\frac{1}{4}}$
 ≥ 9

for $a, b > 0$
 $a+b \geq 2\sqrt{ab}$

$(a+b)^2 \geq 4ab$

$a^2 - 2ab + b^2 \geq 0$

$(a-b)^2 \geq 0$
 true

Q8)

b) i/ from $x = vt \cos \theta$ & $y = vt \sin \theta - \frac{1}{2}gt^2$

$\therefore t = \frac{x}{v \cos \theta}$

$y = v \sin \theta \cdot \frac{x}{v \cos \theta} - \frac{1}{2}g \frac{x^2}{v^2 \cos^2 \theta}$
 $= x \tan \theta - \frac{gx^2 \sec^2 \theta}{2v^2}$

ii/ passes thru (b, h) and (c, h)

$\therefore h = b \tan \theta - \frac{gb^2 \sec^2 \theta}{2v^2}$ (1)

and $h = c \tan \theta - \frac{gc^2 \sec^2 \theta}{2v^2}$ (2)

$\therefore b \tan \theta - \frac{gb^2 \sec^2 \theta}{2v^2} = c \tan \theta - \frac{gc^2 \sec^2 \theta}{2v^2}$

$(b-c) \tan \theta = \frac{(b^2 - c^2)g \sec^2 \theta}{2v^2}$

$\therefore \tan \theta = \frac{(b+c)g \sec^2 \theta}{2v^2}$

$\therefore v^2 = \frac{(b+c)g \sec^2 \theta}{2 \tan \theta}$

iii/ Subst into (1)

$h = b \tan \theta - \frac{gb^2 \sec^2 \theta}{2} \cdot \frac{2 \tan \theta}{(b+c)g \sec^2 \theta}$

$= b \tan \theta - \frac{b^2 \tan \theta}{b+c}$

$h(b+c) = (b^2 + bc) \tan \theta - b^2 \tan \theta$

$= bc \tan \theta$

$\therefore \tan \theta = \frac{h(b+c)}{bc}$